Orhan Kaplan, Ph.D.

Senior Data Scientist | Gaziantep, Turkey | orhankaplan@gantep.edu.tr| +90 5464619339 | https://nalpako2027.github.io/ | https://github.com/nalpako2027?tab=repositories | https://scholar.google.com/citations?user=1q5Kk4UAAAAJ&hl=en

PROFESSIONAL SUMMARY

Senior Data Scientist specializing in causal inference, predictive modeling, and behavioral analytics, with nearly 10 years of research and development experience spanning the University of Florida and Gaziantep University. Experienced in transforming complex, large-scale datasets into rigorous statistical analyses and actionable insights. Skilled in designing observational and quasi-experimental studies, developing predictive models, and estimating causal effects when randomized controlled trials or conventional A/B testing are impractical. Combines advanced statistical reasoning, machine learning, and data visualization to translate complex findings into evidence-based recommendations for strategic decision-making.

TECHNICAL SKILLS

Data Science/Applied Research: Hypothesis Testing, Experimental/Quasi-Experimental Observational Studies Design, Structural Equation Modeling, Propensity Score Matching, Entropy Balancing, Fixed-Effects Modeling, Machine Learning (Multiple Regression, Lasso & Ridge Regression, Logistic Regression, kNN Regression/Classification, Decision-Tree ensembles), Multiple imputation (MI, MICE), and Qualitative Research.

Programming & Development: Languages & Tools (Python, SPSS, Stata, MPlus, Git, SQL (SQLiteStudio, MySQL)), **Machine Learning (**scikit-learn, cross-validation, classification, Random Forest, XGBoost), and **Data & Visualization** (pandas, NumPy, matplotlib/seaborn, Streamlit).

Communication & Collaboration: Strong in rigorous academic/technical writing, team collaboration, supporting a productive and inclusive environment.

Languages: Fluent in English and Turkish.

PROFESSIONAL EXPERIENCE

Lecturer & Researcher | Gaziantep University, *April 2019 – Present*

- Taught graduate-level research methodology and undergrad-level mathematics and teacher education courses.
- Designed and conducted quantitative research studies using SPSS, Stata, Mplus, and Python, applying advanced statistical modeling techniques to educational, psychological, and behavioral datasets.
- Applied propensity score matching, regression-based modeling, mediation and moderation analysis, and other statistical methods to investigate relationships and estimate intervention effects.
- Conducted peer-reviewed research and collaborated on interdisciplinary research projects, contributing to scientific publications and research dissemination.
- Reviewed manuscripts in educational research and quantitative methods, assessing research design, statistical analyses, theoretical frameworks and interpretation of findings.
- As a field editor, evaluated manuscripts and provided editorial recommendations for Dicle University Ziya Gökalp Faculty of Education Journal.

Graduate Research Assistant (Graduate & OPS)|Higher Ed. Dep., University of Florida, *October 2016 – May 2018*

- Managed, cleaned, and prepared large-scale educational datasets for quantitative research and policy evaluation.
- Conducted statistical analyses using Stata, including Regression Discontinuity Design, fixed-effects modeling, and logistic regression.
- Examined state-level educational policies and conducted systematic literature reviews to support research and policy evaluation.

Mathematics Instructor | Upward Bound Program, University of Florida, *January 2012 – May 2012*

- Instructed high school students in ACT-oriented geometry as part of a college preparatory program serving students from low-income backgrounds.

Mathematics &Science Teacher | Batman, Turkey, *September 2006 – February 2008*

- Ikiztepe Middle School (2006-2008) and Sehit Ogretmenler Middle School (2007-2008)

SELECTED PROJECTS

Warehouse Closure Analysis: A SQL-Based Multi-Criteria Decision Framework: A SQL case study using the Mint Classic sample database to determine which warehouse, if any, should be closed, using a two-level weighted scoring model built entirely from queryable data using MySQL Workbench. The decision-making framework visualized with an interactive dashboard using Streamlit and deployed on Streamlit Community Cloud.

MICE-PMM Diagnostics Toolkit: Open-source Python package implementing MICE with Predictive Mean Matching (Rubin, 1987; van Buuren, 2018) for complex survey data, with a diagnostics suite (Rubin's Rules, FMI, RVI, RE, Von Hippel's recommended-M, convergence plots) not previously available together in existing open-source tools.

Telco Customer Churn Analysis & Predictive Modeling Pipeline: End-to-end pipeline identifying high-risk churn segments and simulating retention ROI using logistic regression, KNN, and XGBoost classification, with custom data quality checks and multi-model ROC evaluation.

Hybrid Movie Recommendation System Using Collaborative Filtering & Matrix Factorization: Developed a recommendation pipeline using KNN collaborative filtering, TruncatedSVD, and NMF, with hyperparameter optimization, cross-validation, and comparative model evaluation.

Applications of AI in Mathematics Instruction | TÜBİTAK Grant 222B158 (2023): Instructed mathematics educators on integrating AI-driven tools into instructional design, as part of a nationally funded technological innovation initiative for math education.

EDUCATION

Ph.D. in Curriculum & Instruction, University of Florida, FL, USA *2018*

- Specialization in Mathematics Education and Research and Evaluation Methodology

M.S.E. in Mathematics, Arkansas State University, AR, USA *2011*

- Thesis on *Stone-Weierstrass Approximation Theorem*

Studying Abroad Scholarship, Ministry of National Education, Turkey *October 2007*

- A selective and comprehensive scholarship covering education for Master's and PhD degrees

CERTIFICATIONS

Introduction to Data Science with Python, **HarvardX,** *May 2026*
Machine Learning and AI, HarvardX, *June 2026*
SQL for Data Analysis and Business Insights, Coursera, *August 2026*
Machine Learning and AI Foundations: Causal Inference and Modeling, LinkedIn Learning, *June 2026*
Learning Git and GitHub, LinkedIn Learning, *June 2026*

SELECTED PUBLICATIONS & PRESENTATIONS

Kaplan, O. (2026). Cross-cultural control–value mechanisms on the detrimental effect of bullying on mathematics anxiety. Behavioral Sciences, 16(1), Article 3. https://doi.org/10.3390/bs16010003

Aydın Gürler, S. & Kaplan, O. (2026). The effect of school bullying on science achievement and cognitive appraisals. Problems of Education in the 21st Century, 84(1), 7–26. https://doi.org/10.33225/pec/26.84.07

Kaplan, O., Kacar, T. & Simsek, N. (2025). The effect of bullying on epistemic emotions using Propensity Score Matching: Examining cross-national differences. The paper was presented at 2025 European Conference on Educational Research, Belgrade, Serbia.

Kaplan, O., Kara, M., & Kaçar, T. (2025). The effect of digital technology integration on students' creative thinking skills: PISA evidence using propensity score matching. Problems of Education in the 21st Century, 83(3), 403–423. https://doi.org/10.33225/pec/25.83.403

Aydın Gürler, S. & Kaplan, O. (2023). Attitudes towards STEAM, critical thinking disposition and decision-making skills: Mediation and gender moderation. International Journal of Contemporary Educational Research, 10(1), 210-223. https://doi.org/10.33200/ijcer.1272051

Full publication list available at https://orcid.org/0000-0002-4306-2266 or https://nalpako2027.github.io/about/